Question 41: In an organization, number of departments exists. Each department has a name & unique code. Number of employees work in each department. Each employee has unique employee code. Detailed information like name, address, city, basic, date of join are also stored. In a leave register for each employee leave records are kept showing leave type (CL/EL/ML etc.), from-date and to-date. When an employee retires or resigns then all the leave information pertaining to him are also deleted. Basic salary must be within Rs.5000 to Rs.9000. A department can not be deleted if any employee record refers to it. Valid grades are A/B/C. Employee name must be in uppercase only. Default value for joining date is system date. Design & implement the tables with necessary constraints to support the scenario depicted above.

Query :

```
CREATE TABLE Department(
    dcode VARCHAR(10) PRIMARY KEY,
    dname VARCHAR(50) NOT NULL UNIQUE,
    no_of_employee INT DEFAULT 0
);

CREATE TABLE Employee(
    empcode INT PRIMARY KEY,
    ename VARCHAR(50) NOT NULL,
    address VARCHAR(100),
    city VARCHAR(50),
    basic DECIMAL(8,2),
    doj DATE DEFAULT (CURRENT_DATE),
    grade CHAR(1),
    dcode VARCHAR(10),

    CONSTRAINT chk_basic CHECK (basic BETWEEN 5000 AND 9000),
    CONSTRAINT chk_grade CHECK (grade IN ('A','B','C')),
    CONSTRAINT chk_uppercase CHECK (ename = UPPER(ename)),
    CONSTRAINT fk_emp_dept FOREIGN KEY (dcode) REFERENCES Department(dcode) ON
DELETE RESTRICT
);

CREATE TABLE LeaveRegister(
    leave_id INT AUTO_INCREMENT PRIMARY KEY,
    empcode INT NOT NULL,
    leave_type VARCHAR(10) NOT NULL,
    from_date DATE NOT NULL,
    to_date DATE NOT NULL,

    CONSTRAINT fk_leave_emp FOREIGN KEY(empcode) REFERENCES Employee(empcode)
ON DELETE CASCADE
);
```

Output :

```
mysql> CREATE TABLE Department(
    ->     dcode VARCHAR(10) PRIMARY KEY,
    ->     dname VARCHAR(50) NOT NULL UNIQUE,
    ->     no_of_employee INT DEFAULT 0
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Employee(
    ->     empcode INT PRIMARY KEY,
    ->     ename VARCHAR(50) NOT NULL,
    ->     address VARCHAR(100),
    ->     city VARCHAR(50),
    ->     basic DECIMAL(8,2),
    ->     doj DATE DEFAULT (CURRENT_DATE),
    ->     grade CHAR(1),
    ->     dcode VARCHAR(10),
    ->
    ->     CONSTRAINT chk_basic CHECK (basic BETWEEN 5000 AND 9000),
    ->     CONSTRAINT chk_grade CHECK (grade IN ('A','B','C')),
    ->     CONSTRAINT chk_uppercase CHECK (ename = UPPER(ename)),
    ->     CONSTRAINT fk_emp_dept FOREIGN KEY (dcode) REFERENCES Department(dcode) ON DELETE RESTRICT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE LeaveRegister(
    ->     leave_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     empcode INT NOT NULL,
    ->     leave_type VARCHAR(10) NOT NULL,
    ->     from_date DATE NOT NULL,
    ->     to_date DATE NOT NULL,
    ->
    ->     CONSTRAINT fk_leave_emp FOREIGN KEY(empcode) REFERENCES Employee(empcode) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)
```

Question 42: Try to violate the constraints that you have implemented in the table & note, what happens. [Try with suitable INSERT/UPDATE/DELETE instruction]

Query :

```
INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
(104,'RAKESH','ABC','KOLKATA',10000,'A','D1');

INSERT INTO Employee (empcode,ename,address,city,basic,grade,dcode) VALUES
(105,'RAMESH','ABC','KOLKATA',7000,'D','D1');

INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
(107,'SOUMEN','ABC','KOLKATA',7000,'A','D10');

DELETE FROM Department
WHERE dcode='D1';

DELETE FROM Employee
WHERE empcode=101;
```

Output :

```
mysql> INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
    -> (104,'RAKESH','ABC','KOLKATA',10000,'A','D1');
ERROR 3819 (HY000): Check constraint 'chk_basic' is violated.
```

```
mysql> INSERT INTO Employee (empcode,ename,address,city,basic,grade,dcode) VALUES
    -> (105,'RAMESH','ABC','KOLKATA',7000,'D','D1');
ERROR 3819 (HY000): Check constraint 'chk_grade' is violated.
```

```
mysql> INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
    -> (107,'SOUMEN','ABC','KOLKATA',7000,'A','D10');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOREIGN KEY (`dcode`) REFERENCES
`department` (`dcode`) ON DELETE RESTRICT)
```

```
mysql> DELETE FROM Department
    -> WHERE dcode='D1';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key
constraint fails (`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOR
EIGN KEY (`dcode`) REFERENCES `department` (`dcode`) ON DELETE RESTRICT)
mysql>
```

```
mysql> DELETE FROM Employee
    -> WHERE empcode=101;
Query OK, 1 row affected (0.00 sec)
```

Question 43:

a) create a view showing employee code, name, dcode & Basic For a particular department.

Query :

```
CREATE VIEW D1_EMP_VIEW AS
SELECT empcode, ename, dcode, basic FROM Employee WHERE dcode='D1';
```

Output :

```
mysql> CREATE VIEW D1_EMP_VIEW AS
    -> SELECT empcode, ename, dcode, basic FROM Employee WHERE dcode='D1';
Query OK, 0 rows affected (0.00 sec)
```

b) Try to ensure a row into the view with valid department & also with invalid ones.

Query :

```
INSERT INTO D1_EMP_VIEW VALUES(201,'ANIK','D1',7500);

INSERT INTO D1_EMP_VIEW VALUES(202,'ROHIT','D5',7000);
```

Output :

```
mysql>
mysql> INSERT INTO D1_EMP_VIEW VALUES
    -> (201,'ANIK','D1',7500);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO D1_EMP_VIEW VALUES
    -> (202,'ROHIT','D5',7000);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOREIGN KEY (`dcode`) REFERENCES
`department` (`dcode`) ON DELETE RESTRICT)
```

c) Find the newly inserted row in the table From which view was created .

Query :

```
SELECT * FROM Employee WHERE empcode=201;
```

Output :

```
mysql> SELECT * FROM Employee
    -> WHERE empcode=201;
+---------+-------+---------+------+---------+------+-------+-------+
| empcode | ename | address | city | basic   | doj  | grade | dcode |
+---------+-------+---------+------+---------+------+-------+-------+
|     201 | ANIK  | NULL    | NULL | 7500.00 | NULL | NULL  | D1    |
+---------+-------+---------+------+---------+------+-------+-------+
1 row in set (0.00 sec)
```

d) Try to increment basic by Rs.100/-

Query :

```
UPDATE D1_EMP_VIEW SET basic = basic + 100;
```

Output :

```
mysql> UPDATE D1_EMP_VIEW
    -> SET basic = basic + 100;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

e) Check it in the original table.

Query :

```
SELECT empcode, ename, basic FROM Employee;
```

Output :

```
mysql> SELECT empcode, ename, basic
    -> FROM Employee;
+---------+-------+---------+
| empcode | ename | basic   |
+---------+-------+---------+
|     102 | AMIT  | 6600.00 |
|     103 | SUMAN | 8000.00 |
|     201 | ANIK  | 7600.00 |
+---------+-------+---------+
3 rows in set (0.00 sec)
```

f) Delete the view.

Query :

```
DROP VIEW D1_EMP_VIEW;
```

Output :

```
mysql> DROP VIEW D1_EMP_VIEW;
Query OK, 0 rows affected (0.01 sec)
```

Question 44:
a) create a view Showing empcode, name, deptname, basic, leave type, From date & to date.
Query :

```
CREATE VIEW EMP_LEAVE_VIEW AS
SELECT E.empcode, E.ename, D.dname, E.basic, L.leave_type, L.from_date,
L.to_date
FROM Employee E JOIN Department D ON E.dcode = D.dcode
JOIN LeaveRegister L ON E.empcode = L.empcode;
```

Output :

```
mysql> CREATE VIEW EMP_LEAVE_VIEW AS
    -> SELECT E.empcode, E.ename, D.dname, E.basic, L.leave_type, L.from_date, L.to_date
    -> FROM Employee E JOIN Department D ON E.dcode = D.dcode
    -> JOIN LeaveRegister L ON E.empcode = L.empcode;
Query OK, 0 rows affected (0.03 sec)
```

b) Try to insert a row in the view. Check what happens?
Query :

```
INSERT INTO EMP_LEAVE_VIEW VALUES
(500,'JOY','COMPUTER SCIENCE',7000,'CL','2026-04-01','2026-04-03');
```

Output :

```
mysql> INSERT INTO EMP_LEAVE_VIEW VALUES
    -> (500,'JOY','COMPUTER SCIENCE',7000,'CL','2026-04-01','2026-04-03');
ERROR 1394 (HY000): Can not insert into join view 'assignments.emp_leave_view' without fields list
mysql>
```

c) Try to increment basic by Rs.100.
Query :

```
UPDATE EMP_LEAVE_VIEW
SET basic = basic + 100;
```

Output :

```
mysql> UPDATE EMP_LEAVE_VIEW
    -> SET basic = basic + 100;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

d) Delete the view.
Query :

```
DROP VIEW EMP_LEAVE_VIEW;
```

Output :

```
mysql> DROP VIEW EMP_LEAVE_VIEW;
Query OK, 0 rows affected (0.00 sec)
```

Question 45:

a) Create a table having empcode , Name, deptname, & basic From the existing tables along with the records of the employee who are in a particular department (say, d1) and with a basic Rs. 7000/-

Query :

```
CREATE TABLE D1_EMPLOYEE AS
SELECT E.empcode, E.ename, D.dname, E.basic
FROM Employee E JOIN Department D ON E.dcode = D.dcode
WHERE E.dcode='D1' AND E.basic=7000;
```

Output :

```
mysql> CREATE TABLE D1_EMPLOYEE AS
    -> SELECT E.empcode, E.ename, D.dname, E.basic
    -> FROM Employee E JOIN Department D ON E.dcode = D.dcode
    -> WHERE E.dcode='D1' AND E.basic=7000;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

b) From the existing table, add the employees with the basic salary greater than or equal to 7000/-

Query :

```
INSERT INTO D1_EMPLOYEE
SELECT E.empcode, E.ename, D.dname, E.basic
FROM Employee E JOIN Department D ON E.dcode = D.dcode
WHERE E.basic >= 7000;
```

Output :

```
mysql> INSERT INTO D1_EMPLOYEE
    -> SELECT E.empcode, E.ename, D.dname, E.basic
    -> FROM Employee E JOIN Department D ON E.dcode = D.dcode
    -> WHERE E.basic >= 7000;
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

c) Alter the table to add a net pay column.

Query :

```
ALTER TABLE D1_EMPLOYEE
ADD netpay DECIMAL(10,2);
```

Output :

```
mysql> ALTER TABLE D1_EMPLOYEE
    -> ADD netpay DECIMAL(10,2);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

d) Replace net pay with 1.5* Basic.

Query :

```
UPDATE D1_EMPLOYEE
SET netpay = basic * 1.5;
```

Output :

```
mysql> UPDATE D1_EMPLOYEE
    -> SET netpay = basic * 1.5;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

```
mysql> SELECT * FROM D1_EMPLOYEE;
+---------+-------+------------------+---------+----------+
| empcode | ename | dname            | basic   | netpay   |
+---------+-------+------------------+---------+----------+
|     103 | SUMAN | ELECTRONICS      | 8000.00 | 12000.00 |
|     201 | ANIK  | COMPUTER SCIENCE | 7600.00 | 11400.00 |
+---------+-------+------------------+---------+----------+
2 rows in set (0.00 sec)
```

e) Try to remove the net net pay column. [It may require no. of steps]

Query :

```
ALTER TABLE D1_EMPLOYEE
DROP COLUMN netpay;
```

Output :

```
mysql> ALTER TABLE D1_EMPLOYEE
    -> DROP COLUMN netpay;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Question 46: Drop all the tables that you have created .
Query :

```
DROP TABLE LeaveRegister;

DROP TABLE Employee;

DROP TABLE Department;

DROP TABLE D1_EMPLOYEE;
```

Output :

```
mysql> DROP TABLE LeaveRegister;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP TABLE Employee;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP TABLE Department;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP TABLE D1_EMPLOYEE;
Query OK, 0 rows affected (0.01 sec)
```